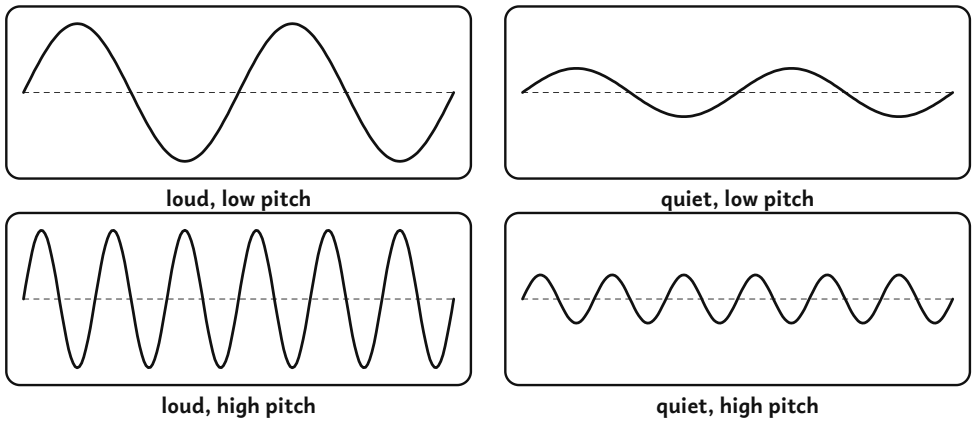
PHYSICAL AND CHEMICAL SCIENCE

Sound

Vibrations, sound waves, pitch and loudness, hearing and noise



In this pack

Learn it: Sound: vibrations that travel as waves ★☆☆
Fact check: Making and hearing sounds ★☆☆
Think it through: Pitch, loudness and waves ★★☆
Investigate: Ruler twang: length and pitch ★★☆
Real-world case study: Too loud! Noise in Dhaka ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

© InquiryMind · Learn · Play · Wonder · Free resources for everyone, bringing quality and equity to education and helping teachers build concepts and make learning visible · Illustrations drawn for InquiryMind · Print in black and white on A4 · Challenge: ★ core ★★ deeper ★★★ stretch

LEARN IT · C5 · SOUND

Sound: vibrations that travel as waves

Name: Class: Date:

Learning goals

- Explain that sounds are made by vibrations and travel as waves.
- Use a wave model to explain pitch and loudness.
- Describe how we hear and how to protect our hearing.

Tap a dhol, pluck an ektara string, or put your fingers on your throat and hum. You will feel something moving quickly backwards and forwards. These tiny, fast movements are **vibrations**, and every sound starts with one.

How sound travels

A vibrating object pushes on the particles of the air next to it. They push the next particles, and the next, passing the vibration along like a ripple. This is a **sound wave**. Sound can travel through **gases** (air), **liquids** (water) and **solids** (wood, metal, string). It cannot travel through **empty space**, because there are no particles to pass the vibration on. In space, no one can hear you shout!

Material	Speed of sound (approximate)
air	340 metres per second
water	1500 metres per second
steel	5900 metres per second

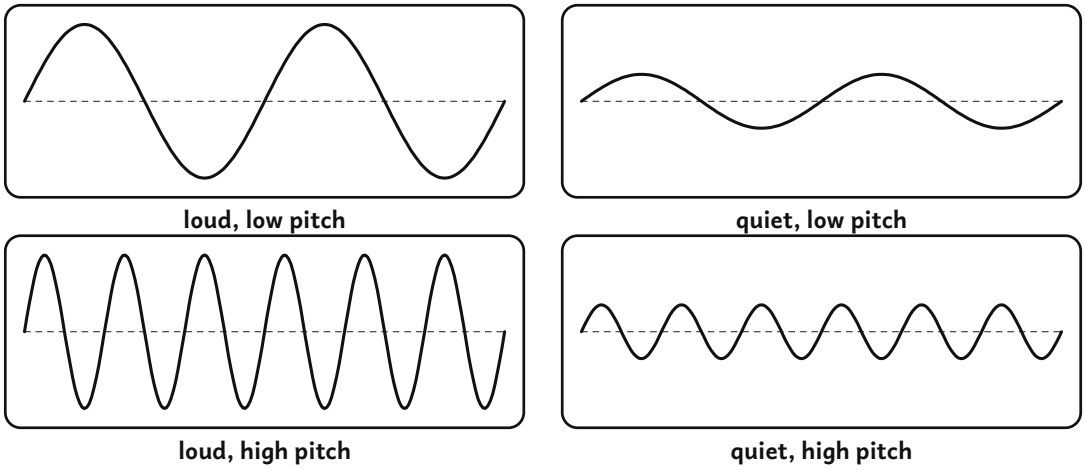
Sound travels faster through solids and liquids because their particles are closer together. Light travels much faster than sound, which is why we see lightning before we hear the thunder.

Pitch: high or low

Pitch is how high or low a sound is. Fast vibrations make a **high** pitch; slow vibrations make a **low** pitch. The number of vibrations each second is called the **frequency**, measured in hertz (Hz). On a string instrument, a **shorter**, **tighter** or **thinner** string gives a higher pitch. On a flute (*bansuri*), covering holes makes the column of air longer and the pitch lower.

Loudness: loud or quiet

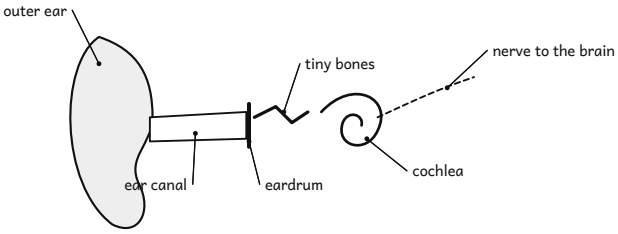
Loudness depends on how big the vibrations are. Hitting a drum harder makes bigger vibrations and a **louder** sound. On a wave diagram, the **height** of the wave shows loudness and how **close together** the waves are shows pitch.



Comparing sound waves.

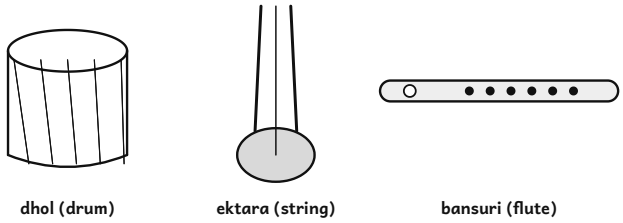
How we hear

The **outer ear** collects sound and funnels it down the **ear canal** to the **eardrum**, which vibrates. Three **tiny bones** pass the vibrations to the **cochlea**, which turns them into messages that travel along a **nerve** to the **brain**.



Protect your hearing

Loudness is measured in **decibels (dB)**. A whisper is about 30 dB and normal talking about 60 dB. Listening to sounds louder than about 85 dB for a long time can **permanently damage** hearing. Keep headphones turned down and cover your ears near very loud noises.



Bangladeshi instruments: a drum, a string instrument and a wind instrument.

Key words

vibration	a quick movement backwards and forwards
sound wave	vibrations passed from particle to particle
pitch	how high or low a sound is
frequency	number of vibrations each second (Hz)
loudness	how loud or quiet a sound is
decibel (dB)	unit for measuring loudness
eardrum	thin skin in the ear that vibrates when sound reaches it

FACT CHECK · C5 · SOUND

Making and hearing sounds

★☆☆
Grade 5

Name: Class: Date:

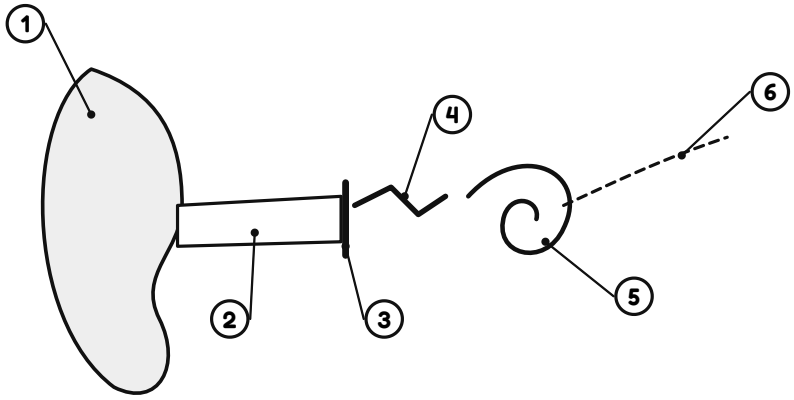
Learning goals

- Recall how sound is made and heard.
- Label the ear.

1 Label the parts of the ear. [6]

LABELS

eardrum **nerve to the brain** **ear canal** **tiny bones** **outer ear** **cochlea**



1 _____ 2 _____ 3 _____

4 _____ 5 _____ 6 _____

2 Fill in the gaps. [5]

WORD BANK

low **vibrations** **wave** **high** **space** **light** **gases**

All sounds are made by _____. Sound travels as a _____. It can travel through solids, liquids and _____, but not through empty _____. Fast vibrations make a _____ pitch.

3 What vibrates to make the sound? [4]

Instrument	What vibrates?
dhol (drum)	
ektara	
bansuri (flute)	
your voice	

4 **True or false?** [4]

Statement	True	False	If false, correct it
1. Sound travels faster through steel than through air.			
2. A bigger vibration makes a quieter sound.			
3. Astronauts on the Moon can hear each other without radios.			
4. A shorter string makes a higher pitch.			

Pitch, loudness and waves

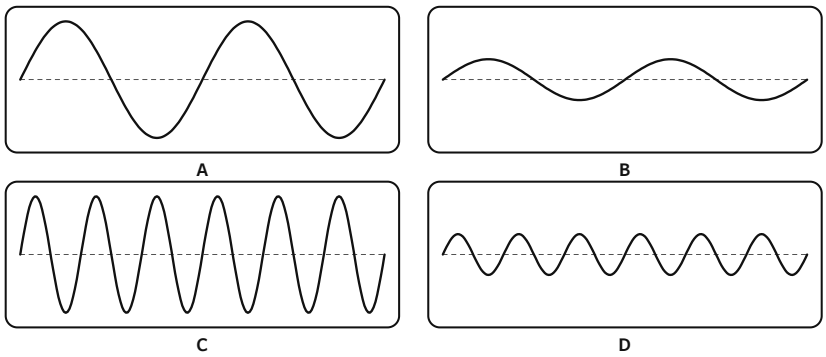
★★☆ Grade 5

Name: Class: Date:

Learning goals

- Use wave diagrams.
- Explain pitch and loudness changes.

1 **Look at the four waves A to D.** [4]



a) Which wave is the loudest with the lowest pitch?

.....

b) Which wave is the quietest with the highest pitch?

.....

c) What is the same about waves A and C? What is different?

.....

.....

2 **An ektara player squeezes the bamboo neck to make the string tighter. What happens to the pitch? Why?** [2]

.....

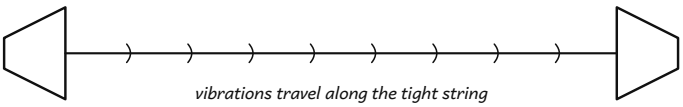
.....

3 **Why do we see lightning before we hear the thunder?** [2]

.....

.....

4 **A string telephone works only when the string is tight. Explain.** [2]



vibrations travel along the tight string

INVESTIGATE · C5 · SOUND

Ruler twang: length and pitch

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Investigate pitch with a fair test.
- Plot and use a line graph.
- Explain using vibrations.

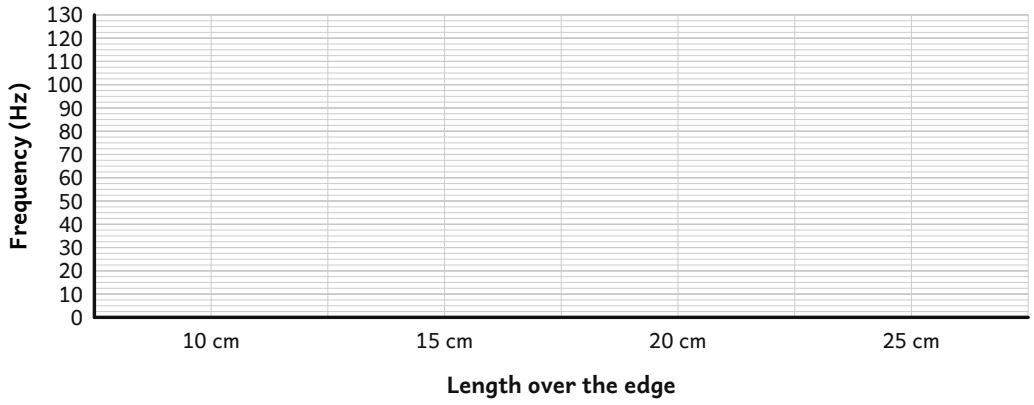
Tania held a ruler on the edge of a table and twanged the end. She changed how much of the ruler stuck out over the edge. A sound app on a tablet counted the vibrations each second (frequency).

1 **Name the variables.** [3]

Change	
Measure	
Keep the same	

2 **Plot the results as a line graph.** [4]

Length over the edge (cm)	10	15	20	25
Frequency (Hz)	120	55	30	20



3 **Describe the pattern. Which length made the highest pitch?** [2]

.....

.....

4 **Predict the frequency for 12 cm.** [1]

.....

5 **How could Tania test how loudness changes? Write a plan in one sentence.** [2]

.....

.....

REAL-WORLD CASE STUDY · C5 · SOUND

Too loud! Noise in Dhaka

★★★ Grade 5

Name: **Class:** **Date:**

Learning goals

- Use data about loudness.
- Explain health effects of noise.
- Evaluate solutions from different viewpoints.

Science | Health | Maths | Citizenship | Ethics

In 2022, a United Nations Environment Programme report named **Dhaka** as one of the noisiest cities in the world. Car and bus horns, construction machines, loudspeakers and generators fill the streets with sound.

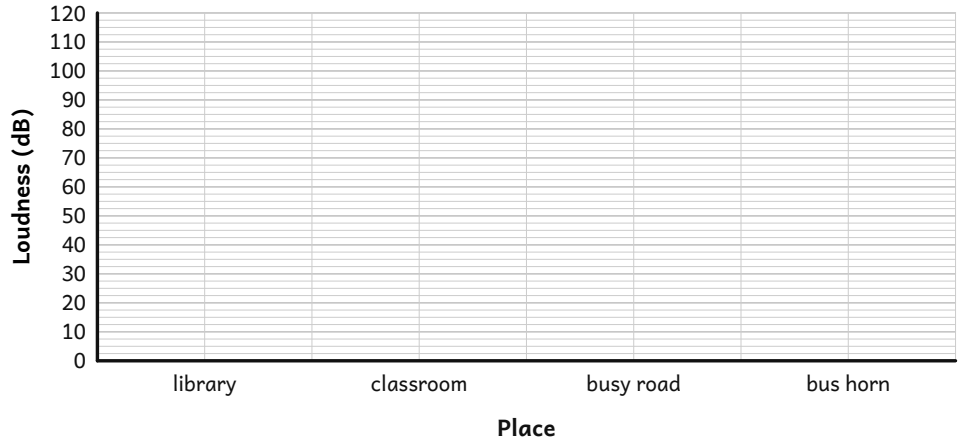
Doctors say that living with loud noise every day can cause **hearing loss**, headaches, stress, high blood pressure and poor sleep. Children in noisy schools find it harder to concentrate and learn.

Bangladesh has **noise rules**: areas around hospitals, schools and offices are meant to be **silent zones**, with limits as low as 50 dB in the daytime, and very loud hydraulic horns are banned. But the rules are often broken.

Sources: UNEP Frontiers Report 2022; Bangladesh Noise Pollution (Control) Rules 2006.

1 **A class measured the loudness at different places with a sound meter app (example readings).** [4]

Place	school library	classroom	busy road outside school	bus horn nearby
Loudness (dB)	35	60	85	110



a) Draw the bar chart.

b) Which places are louder than the 50 dB limit for a silent zone?

c) Which place could damage hearing if people stayed there for a long time?

2 **Explain why a loud horn could harm the ear, using what you know about the eardrum.** [2]

3 **What might each person say?** [3]

Person	View
a bus driver	
a teacher near the road	
a traffic police officer	

4 **Design a "Quiet School Zone" plan with three actions.** [3]

REVIEW AND CHALLENGE · C5 · SOUND

Show what you know

★★★
Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

H	E	U	A	Y	S	T	R	I	N	G
A	C	B	T	L	O	U	D	Y	C	P
D	H	N	S	R	R	F	L	S	V	E
J	O	Y	F	V	M	G	T	V	E	F
K	Y	J	V	L	F	I	D	F	A	Y
W	J	F	I	D	U	T	E	F	R	C
A	W	J	B	R	U	P	C	I	D	C
V	O	S	R	U	F	I	I	B	R	G
E	M	O	A	M	K	T	B	A	U	Y
A	U	M	T	I	S	C	E	A	M	I
J	A	E	E	N	I	H	L	K	O	G

decibel **drum** **eardrum** **echo** **loud** **pitch** **string** **vibrate** **wave**

2 **Which change makes a guitar string sound higher?** [1]

- ◯ make it longer
- ◯ make it looser
- ◯ make it shorter
- ◯ pluck it more gently

3 **Why can't sound travel in space?** [1]

- ◯ it is too cold
- ◯ there are no particles to pass on vibrations
- ◯ it is too dark
- ◯ the Sun is too loud

4 **Draw two waves: one quiet, high-pitched sound and one loud, low-pitched sound.** [2]

quiet, high pitch

loud, low pitch

5 **Challenge. Whales can hear each other across hundreds of kilometres of ocean. Use what you know about sound to suggest why.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain that sounds are made by vibrations	○	○	○
explain how sound travels through solids, liquids and gases	○	○	○
use wave diagrams to compare pitch and loudness	○	○	○
explain how we hear	○	○	○
investigate how length changes pitch	○	○	○
explain why noise is a health problem and suggest solutions	○	○	○